Competition Guide

for the

Cortical Peaks Challenge 2026

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General Information

The Cortical Peaks Challenge is a brain-computer interface (BCI) competition held as part of the 10th Graz BCI Conference. The challenge provides a platform for teams to evaluate and demonstrate their BCI systems in a series of standardized gaming tasks that require discrete and continuous control.

The competition aims to foster innovation in BCI research, encourage the exchange of ideas between research groups, and provide an engaging environment for benchmarking non-invasive BCI technologies under realistic conditions. Teams are invited to develop and optimize their systems to achieve robust and intuitive control across multiple game scenarios.

For game control, any type of non-invasive brain-computer interface (BCI) is permitted. The term BCI is used according to the working definition of the BCI Society [1]. Consequently, the intentional use of physiological artifacts, such as eye blinks, jaw clenching, or other voluntary muscular activity, to improve classification performance is not permitted.

The challenge is open to teams competing with pilots with spinal cord injury (SCI) as well as non-disabled pilots. By including both participant groups, the competition aims to promote the development and evaluation of BCI technologies across diverse user populations.

[1] BCI Society. Definition of Brain-Computer Interface. https://bcisociety.org/bci-definition

Eligibility

Technical Eligibility

TE-1 Only non-invasive brain-computer interface (BCI) systems are permitted for use in the Cortical Peaks Challenge. All hardware components used during the competition must be commercially available at the time of the event.

TE-2 Control signals must originate exclusively from brain activity. The intentional use of physiological artifacts, including but not limited to eye movements, eye blinks, facial muscle activity, jaw contractions, teeth clenching, or other voluntary muscular actions, as a means of game control is strictly prohibited.

Important: Competition officials will monitor pilots and technical systems throughout the competition to verify compliance with the eligibility and technical requirements. Particular attention will be given to the detection of prohibited control signals originating from muscle activity, eye movements, or other non-neural sources.

TE-3 Signal processing pipelines must include artifact detection and removal procedures. Physiological artifacts, including but not limited to eye movements, eye blinks, facial muscle activity, jaw contractions, and teeth clenching, must be removed prior to the classification or regression stage. Such artifacts must not be used, directly or indirectly, as features for classification, regression, or any other control mechanism. Any system found to intentionally exploit physiological artifacts for control purposes will be disqualified from the competition.

TE-4 Visual displays may be used to present game information and user feedback to the pilot. Additional monitors are permitted and may either be mounted to the wheelchair or placed on a table directly in front of the pilot. However, these displays must not be used to elicit control signals through externally evoked responses such as steady-state visual evoked potentials (SSVEPs), code-modulated visual evoked potentials (c-VEPs), or similar paradigms. The competition is intended to evaluate self-generated brain activity rather than responses induced by external stimulation.

TE-5 All teams are required to use the official UDP communication interface provided by the organizers. The communication protocol and message structure must be implemented exactly as specified in the technical documentation. Modifications to the game software, communication protocol, or any other component of the organizer-provided interface are not permitted. All game control signals must be transmitted exclusively through the designated UDP interface.

Pilot Eligibility

PE-1 The SCI category is intended for individuals with cervical spinal cord injuries that result in substantial motor impairment. Eligibility will be assessed based on the International Standards for Neurological Classification of Spinal Cord Injury (ISNCSCI). As a general requirement, pilots should have a neurological level of injury at C5 or above and an AIS classification of A, B, or C. The organizers reserve the right to review medical documentation to confirm eligibility.

Games

Game-BCI Connection via UDP

1. System Architecture

The framework operates on a server-authoritative, headless game engine (built via Pygame) over **UDP**. The server operator retains exclusive control over starting and stopping matches; clients cannot initiate or terminate game instances. The participants connect their BCI controller as a client to the server. No game state will be shared with the BCI controller (client) while the games are being played.

2. Data Representation & Encoding

- **Strings:** Encoded as UTF-8 inside fixed-size byte arrays. Shorter strings must use trailing null padding (0x00). On decode, trailing nulls must be stripped. An all-null field denotes an empty string.
- **Session Tokens:** Transmitted as 8 raw bytes. Represented textually as 16-character lowercase hexadecimal strings.
- **Byte Offsets:** Zero-indexed from the absolute beginning of the UDP datagram.
- **Field Value Reference Table**:
    - **Type Identifiers**:
        - 0x01 = REGISTER,
        - 0x02 = ACK
        - 0x03 = PING
        - 0x04 = PONG
        - 0x05 = CMD

All multi-byte fields are big-endian. On little-endian hosts, byte-swap these fields on both read and write. All messages have statically known or easily computed sizes. A BCI client only ever receives ACK (9 bytes), server PING (1 byte), and server PONG (1 byte). A receive buffer of 32 bytes is fully sufficient.

Example for **registering**:

```
01  00  61 6c 69 63 65 00 00 00 00 00 00 00 00 00 00 00
^   ^   ^-- "alice" (5 bytes) --^  ^--- zero padding ---^
|   device=BCI
type=REGISTER
```

Example for sending a **ping** to the server:

```
03  00  ab cd ef 01 23 45 67 89
^   ^   ^--- session_token (8 bytes) ---^
|   device=BCI
type=PING
```

Example for confirmation of connection from server and receiving the session token:

```
02  ab cd ef 01 23 45 67 89
^   ^--- session_token (8 bytes) ---^
type=ACK
```

Example for **sending a command**:

```
05  00  ab cd ef 01 23 45 67 89  6a 75 6d 70 00 00 00 00
^   ^   ^--- session_token ---^  ^-"jump"-^  ^-padding-^
|   device=BCI
type=CMD
```

3. Connection & Game Lifecycle

- **Session Initialization:** The client sends a REGISTER message to receive a unique, non-reclaimable session token.
- **Heartbeat Mechanism:** Operates bidirectionally with a strict **3-second inactivity timeout**.
  1. *Client-to-Server:* The client sends a 10-byte PING. The server responds with a 1-byte PONG. Any valid incoming server message resets the client's timer. If 3 seconds elapse without data, the client discards its token and resumes registration.
  2. *Server-to-Client:* The server sends a 1-byte PING. The client responds with a 10-byte PONG. Any valid client transmission updates the server's timestamp.
- **Disconnection & Eviction:** Missing client data for >3 seconds flags the client as disconnected and pauses the game. Any valid client message containing the session token within a 30-second window forces an immediate reconnection. After 30 seconds of total silence, the session is evicted, requiring a new registration.
- **State Machine Transitions:**
  1. *Countdown Phase:* Game is initialized but paused. Incoming CMD messages are discarded.
  2. *Active Phase:* Countdown reaches zero. Game loops process incoming CMD inputs.
  3. *Linger Phase:* Triggered by game over (obstacle collision in Brainski 1; score limit reached in Pong). The server holds the final frame and blocks commands before resetting.

4. Input Mappings

The game framework accepts 4 different control signals sent via the CMD message payload. These are mapped contextually across different games:

- INPUT_A: Maps to Brainski 1 (Jump), Brainski 2(Left Rotation)
- INPUT_B: Maps to Brainski 2 (Duck), Brainski 2(Right Rotation)
- INPUT_C: Maps to Pong (Move Up), Brainski 2(Forward)
- INPUT_D: Maps to Pong (Move Down), Brainski 2(Backward)

Clients deliver these by transmitting the exact string identifier (e.g., "INPUT_A" or "INPUT_B") zero-padded to 8 bytes inside the command packet, which the server translates based on the active game.

Disclaimer: All the information is subject to minor changes, but the general structure will stay the same. A more detailed implementation document, as well as an example controller implementation will be provided to participants in the future.

Game Description

The competition consists of four game variants: BrainSki 1 (Jump Only), BrainSki 1 (Jump & Duck), BrainSki 2 (Static Obstacles), and BrainSki 2 (Dynamic Obstacles). Each game is played twice, with the order randomized for each team.

Detailed descriptions of the individual games are provided below.

BrainSki 1

BrainSki 1 is an obstacle-avoidance game in which the player controls an in-game character using discrete BCI commands. Depending on the game variant, the player must either jump over obstacles or both jump over and duck under obstacles. The obstacles automatically move toward the player and require no additional control input.

Each obstacle is preceded by a gray horizontal marker indicating the optimal command window. Issuing the correct command while the obstacle is within this window allows the player to successfully pass the obstacle. Commands issued too late will result in a collision. Commands issued too early may still result in a successful pass, provided that the character returns to the appropriate position before reaching the obstacle. Issuing an incorrect command during the valid command window will result in a collision.

Two BrainSki 1 variants are included in the competition:

- BrainSki 1 (Jump Only): Players use a single control command to jump over obstacles.
- BrainSki 1 (Jump & Duck): Players use two control commands to either jump over or duck under obstacles.

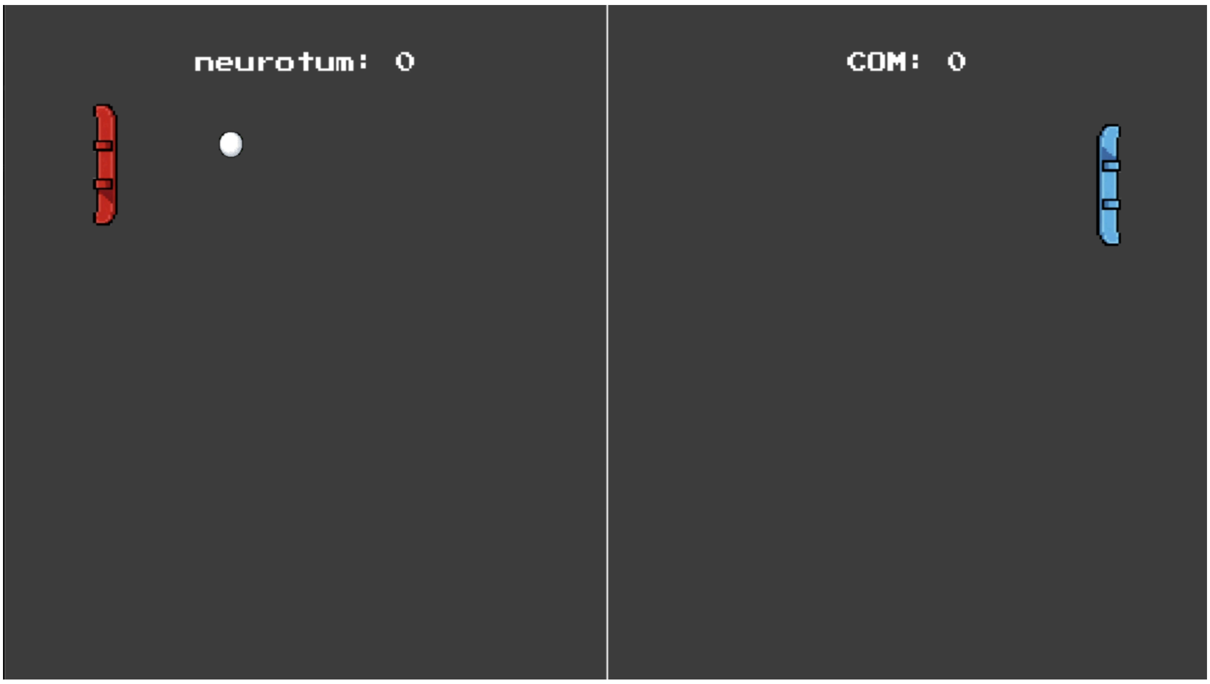
BrainSki 2

A detailed description of BrainSki 2 will be provided in a future revision of this document.

Pong

Depending on the final number of participating teams, a BCI-controlled version of the classic Pong game may be used as a tie-breaker or exhibition match. For example, the game may be played between the two highest-ranked teams at the end of the competition.

Whether Pong will contribute to the overall competition ranking or be conducted solely as a showcase event will be announced prior to the start of the competition.



Scoring

BrainSki 1

Each BrainSki 1 run has a maximum duration of 3 minutes. Players start the game with three lives. A life is lost whenever the player collides with an obstacle. No additional lives can be obtained during gameplay.

The objective is to survive for as long as possible. If a player loses all three lives before the 3-minute time limit expires, the run ends immediately and the achieved survival time is recorded as the score. If the player successfully survives until the end of the 3-minute run, the player receives the maximum base score. Any remaining lives are awarded as additional bonus points.

To ensure fairness across participants, each run contains an equal number of jump and duck obstacles. The order of these obstacles is generated using a seeded randomization procedure and is identical for all competitors.

These values and game parameters may be subject to minor adjustments based on participant feedback and final testing prior to the competition.

BrainSki 2

In BrainSki2 the time needed to reach the target is used as a scoring metric. A more detailed scoring description is provided once the game is published.

Pong

Unlike the other games, Pong is played directly between two competing pilots. Each player starts the match with a fixed number of lives. The match ends when one player has lost all of their lives, at which point the opposing player is declared the winner.

On-Site Information and Technical Requirements

1. **What will be provided by the organizers?**

   The organizers will provide the following infrastructure for all participating teams:

   - Power outlets and extension cords
   - One external 27-inch monitor including an HDMI connection
   - Competition tables and seating
   - Dedicated rooms for EEG preparation, system setup, and training runs prior to the competition

   Additional information regarding the venue and available facilities will be shared closer to the event.

2. **What do teams need to bring?**

   Teams are responsible for bringing all equipment required to operate their BCI system, including:

   - Non-invasive BCI hardware (e.g., amplifier, electrodes, electrode cap)
   - A laptop or workstation running the required acquisition, processing, and control software
   - All necessary cables, adapters, chargers, and power supplies
   - Consumables required for EEG preparation (e.g., gel, saline solution, cleaning materials)
   - Any additional equipment required for setup, calibration, and operation of the BCI system

   Teams are encouraged to bring spare components whenever possible to ensure continued operation in the event of equipment failure.

3. **What are the technical requirements for their setup?**

   The competition laptop must provide an Ethernet network connection, either through a built-in Ethernet port or a suitable adapter. This connection is required for communication with the competition games via the official UDP interface.

In addition, teams should be able to execute Python scripts on their competition laptop. Python is only required for testing and validation purposes prior to the competition and is not required for gameplay itself.

4. **What happens on competition day?**

   The competition will take place in a large lecture hall as part of the Graz BCI Conference. During each match, two teams from the same category (non-disabled or SCI) will compete simultaneously.

   The game screens of the competing teams will be displayed alternately on a large projection screen visible to the audience, allowing attendees to follow the competition in real time. The possibility of having a live moderator to guide the audience through the event is currently under discussion.

   The final competition format will depend on the number of registered teams and will be announced once registration has closed.

Fair Play and Competition Integrity

The Cortical Peaks Challenge is intended as a community-driven event that provides researchers, developers, and pilots with an opportunity to demonstrate and compare their BCI systems in a competitive yet collaborative environment.

In contrast to large-scale international competitions, the organizers do not have the resources to perform comprehensive technical audits of participating systems or detailed medical verification of pilot eligibility. Consequently, compliance with the competition rules relies to a large extent on the honesty and integrity of all participants.

By registering for the competition, teams confirm that their submitted systems comply with the technical requirements described in this guide and that pilots meet the eligibility criteria for their respective competition category.

The organizers reserve the right to observe participants, request clarifications regarding technical implementations, and investigate potential rule violations when reasonable concerns arise. However, the competition is fundamentally based on mutual trust, scientific integrity, and fair play.

We ask all teams to respect both the letter and the spirit of the competition rules. The goal of the Cortical Peaks Challenge is not only to identify high-performing BCI systems, but also to promote scientific exchange, innovation, and collaboration within the BCI community.